

Complex-Weighted Operational Theories and Quantum Algorithms

Complex-weighted operational semantics. Suppose that the weighting construction for an operational theory is specialized to the weight object $\Omega = \mathbb{C}$. Let $\text{Proc} = \mathcal{T}(1, P)$ be the set of closed processes and let $\mathcal{H}$ be the Hilbert space with orthonormal basis $\{|p\rangle : p \in \text{Proc}\}$. A complex-valued weight map on rewrites induces the operator

$$A|p\rangle = \sum_{r:s(r)=p} \omega(r)|t(r)\rangle.$$

If several rewrites have the same source and target, their amplitudes add. Thus the quiver structure of the operational theory becomes a complex-weighted adjacency operator. Local summability guarantees that the displayed sum is well defined on basis vectors, while boundedness of A is a separate condition needed to extend it to all of $\mathcal{H}$.

If $\|A\| \leq 1$, the contraction A has the standard unitary dilation on $\mathcal{H} \oplus \mathcal{H}$,

$$U = \begin{pmatrix} A & (I - AA^*)^{1/2} \\ (I - A^*A)^{1/2} & -A^* \end{pmatrix}.$$

For an arbitrary bounded A , one first replaces it by A/α for $\alpha \geq \|A\|$. Identifying $\mathcal{H} \oplus \mathcal{H}$ with $\mathcal{H} \otimes \mathbb{C}^2$, applying the dilation with a fresh ancillary qubit at each step and postselecting every ancilla onto $|0\rangle$ realizes A^n/α^n after n steps. When the weighted rewrite operator is already unitary, no dilation or postselection is required.

Direct encoding of Grover search

Let X be a finite search type with N closed values, and introduce two families of process constructors

$$Q_O, Q_D : X \rightarrow P.$$

The two families distinguish the oracle and diffusion phases of a Grover iteration. Let $f : X \rightarrow \{0, 1\}$ be the search predicate. The oracle is represented by the rewrite family

$$o(x) : Q_O(x) \longrightarrow Q_D(x), \quad \omega(o(x)) = (-1)^{f(x)}.$$

Consequently the induced map from the Q_O sector to the Q_D sector is the usual phase oracle

$$O_f|x\rangle = (-1)^{f(x)}|x\rangle.$$

The diffusion operator has an equally direct quiver presentation. Introduce a rewrite

$$d(x, y) : Q_D(x) \longrightarrow Q_O(y)$$

for every pair $x, y \in X$, with

$$\omega(d(x, y)) = \frac{2}{N} - \delta_{xy}.$$

For a fixed source x , there are therefore N outgoing rewrites: amplitude $2/N$ to every distinct y , and amplitude $2/N - 1$ on the self-loop. The induced operator is

$$D|x\rangle = \sum_{y \in X} \left(\frac{2}{N} - \delta_{xy} \right) |y\rangle = \frac{2}{N} \sum_{y \in X} |y\rangle - |x\rangle.$$

Writing

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{y \in X} |y\rangle,$$

we obtain

$$D = 2|s\rangle\langle s| - I,$$

which is exactly Grover's diffusion operator.

In a pi-calculus-like syntax with internal complex-valued **where** expressions, the rule can be presented schematically as

$$\mathbf{x}, \mathbf{y} : X \vdash \text{diffuse}(\mathbf{x}, \mathbf{y}) : Q_D(\mathbf{x}) \rightsquigarrow Q_D(\mathbf{y}) \text{ where } 2/N - \text{delta}(\mathbf{x}, \mathbf{y}).$$

The significant feature is that y is a finite parameter of the rewrite rule that is not determined by matching its source. Its ground instances generate the complete family of outgoing edges from $Q_D(x)$. This use of a target-only parameter is sufficient to express dense linear operators.

On the Hilbert space $\mathcal{H}_O \oplus \mathcal{H}_D$ spanned by the two process families, one small operational step has block form

$$A = \begin{pmatrix} 0 & D \\ O_f & 0 \end{pmatrix}.$$

Since D and O_f are unitary, so is A , and

$$A^2 = \begin{pmatrix} DO_f & 0 \\ 0 & O_f D \end{pmatrix}.$$

Thus, starting in the oracle sector, two small-step reductions implement one Grover iterate. After approximately $\frac{\pi}{4}\sqrt{N/M}$ iterations, where M values satisfy f , measurement has the usual Grover success probability. The initial uniform state may be taken as the initial state of the semantics, or prepared by the familiar Hadamard construction when N is a power of two.

Encoding Shor period finding

The same mechanism is expressive enough for Shor's algorithm. Let $Q = 2^m$ be the size of the exponent register and let N be the integer to be factored. For an integer a relatively prime to N , use process constructors whose basis states represent pairs

$$|x\rangle|z\rangle, \quad 0 \leq x < Q, \quad 0 \leq z < N.$$

As with Grover, distinct process families can mark successive phases of the algorithm, ensuring that only the rewrite rules for the current phase are enabled.

Hadamard rewrite rules prepare

$$\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle|1\rangle.$$

Modular exponentiation can then be represented directly by the deterministic rewrite family

$$m_a(x, z) : Q_M(x, z) \longrightarrow Q_F(x, za^x \bmod N), \quad \omega(m_a(x, z)) = 1.$$

Because a is invertible modulo N , multiplication by a^x permutes the second register for each x , so this rewrite family induces a unitary permutation. Starting with $z = 1$, it produces the standard state

$$\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle|a^x \bmod N\rangle.$$

The inverse quantum Fourier transform on the first register also has a direct weighted-quiver encoding. Introduce

$$q(x, y, z) : Q_F(x, z) \longrightarrow Q_R(y, z)$$

with

$$\omega(q(x, y, z)) = \frac{1}{\sqrt{Q}} \exp\left(-\frac{2\pi i xy}{Q}\right).$$

For each basis process $Q_F(x, z)$ this creates Q outgoing rewrites, and the induced operator is exactly $F_Q^{-1} \otimes I$. Measurement of the first register followed by the usual classical continued-fraction computation recovers the period with the same analysis as in Shor's algorithm.

This direct Fourier rule establishes expressivity but is exponentially dense in the number m of exponent qubits. A succinct operational theory can instead decompose it into Hadamard and controlled-phase rewrite families, requiring only polynomially many rule applications. Modular exponentiation can likewise be decomposed into reversible controlled modular multiplications. Thus the operational-theory formalism supports both a direct matrix-level encoding of Shor's quantum operators and the usual polynomial-size circuit decomposition. The role of complex-weighted rewriting is to provide the linear semantics; efficiency depends, as in the circuit model, on whether the required rewrite families admit succinct and efficiently realizable presentations.